Chan Young Joe

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Current Position

Researcher, Data Science Team — Jan. 2026 – Present
Research Institute of Environmental Health and Safety, EH R&C Co. — *Incheon, Korea*

Education

University of Seoul — Seoul, Korea
M.S. in Applied Chemistry (Computational Chemistry), advisor Prof. Rakwoo Chang — *Sep. 2024 – Feb. 2026*

University of Seoul — Seoul, Korea
B.S. in Applied Chemistry; B.E. in Artificial Intelligence (double major) — *Mar. 2021 – Aug. 2024*

National Institute for Lifelong Education (NILE), Korea University — Seoul, Korea
B.S. in Mathematics — *Jan. 2025 – Dec. 2026 (expected)*

Research Experience

Researcher, Computational Molecular Modeling Laboratory — Jun. 2021 – Feb. 2026
University of Seoul (Prof. Rakwoo Chang) — *Seoul, Korea*

- Large language models (in-context learning, fine-tuning) for molecular property prediction
- Molecular dynamics of membranes and enzymes; hybrid multigrained sampling

Visiting Researcher, Department of Bioengineering — Jun. 2024 – Aug. 2024
Lehigh University (Prof. Wonpil Im) — *Bethlehem, PA, USA*

Selected Publications

Lee, J.-H.*; **Joe**, **C. Y.** A flux-based bioaccumulation model for growing organisms: integrating advective–diffusive uptake at exchange organs into dynamic energy budget theory. *Environ. Toxicol. Chem.* **2026**, vgag253, DOI: 10.1093/etojnl/vgag253.

Joe, **C. Y.**; Song, K.*; Chang, R.* Evaluating in-context learning in large language models for molecular property regression. *J. Comput. Chem.* **2026**, *47*, e70308, DOI: 10.1002/jcc.70308.

Open-Source Software

PFAS–rice uptake model | *Python, HYDRUS-1D, Streamlit* — github.com/ccy5123/pfas-rice-model

- Mechanistic four-compartment (root–stem–leaf–grain) uptake model for anionic PFAS in paddy rice, covering 12 calibrated PFCA/PFSA congeners
- Coupled to HYDRUS-1D soil transport; parameterisable from chemical structure; interactive web application

sci-adk | *Python, Claude Code* — github.com/ccy5123/sci-adk

- Verification kit for agent-assisted research: AI agents propose, and a deterministic engine judges each claim against decision rules frozen before the experiment
- Append-only evidence record with headless re-verification by third parties and tamper-evident provenance

Selected Presentations

Joe, **C. Y.**; Lee, J.-H.* Integrated DEB-TK model for fish bioaccumulation assessment of neutral organic chemicals: theoretical formulation and empirical validation. Presented at the 2026 Spring General Meeting of the Korean Society of Environmental Health, Jeju, Korea, May 2026; poster P-EaRA-40. Best Poster Award.

Joe, **C. Y.**; Song, K.*; Chang, R.* Evaluating in-context learning in large language models for molecular property regression. Presented at Pacifichem 2025, Honolulu, HI, Dec. 15–20, 2025; poster PHY006.

*Corresponding author

Honors

Best Poster Award, 2026 Spring General Meeting of the Korean Society of Environmental Health	2026
Outstanding Paper Award, President of the University of Seoul (*J. Comput. Chem.* paper)	2026
Best Poster Award, Winter Workshop of the Global Science Research Center for Systems Chemistry	2025
Best Poster Award, Fall Meeting of the Korean Society of Environmental Health and Toxicology	2024
President's Award, graduation with honors (cum laude), University of Seoul	2024